LAB 01

Pipeline Skeleton, Configuration and the Alpaca Paper Account

Item	Detail
Topic	Topic 1: Foundations of Quant and Algorithmic Trading: From Thorp to AI Agents
Pipeline stage	Configuration
Duration	20 min
Simulator page	http://localhost:8000/#/settings
Mode	No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

Lion City Capital (fictitious), a Singapore-based systematic trading desk, is building an automated algorithmic trading pipeline for US equities and ETFs on an Alpaca paper-trading account. Every later stage - data, signals, execution, automation - reads one configuration file, so the desk starts by creating that file, keeping the API keys outside the code, and proving that the paper account, the market clock and the data path answer.

Objective

Get the simulator running, decide between the offline demo and an Alpaca paper account, and check every integration's status.

Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 01. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - you will use it in the debrief and in the later labs.

Lab Steps - In the Trading Simulator

No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 01: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

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Step 1. Load the demo data

Open Settings and press 'Load / refresh demo data'. This caches 12 symbols of daily bars 2018-2025, SPY hourly bars, sample headlines, two custom strategies and sample backtests.

Open: http://localhost:8000/#/settings (Settings > Demo data)

Check: Settings shows about 29,000 bars and at least 8 backtests.

Step 2. Check the integrations

Open Integrations. Without keys, Alpaca shows 'not configured' and the app uses the simulator broker and the offline data. To connect Alpaca paper: alpaca.markets > Paper Trading > API Keys > Generate, then paste the Key ID and Secret into the Alpaca card and press Save & test.

Open: http://localhost:8000/#/integrations (Integrations)

Check: Alpaca test says 'Paper account ACTIVE' (online) - or you understand the app runs fully offline.

Step 3. Check the AI agent login

On the Integrations page read the OpenAI Codex and Claude cards. They are optional: without them the multi-agent workflow runs its rule-based agents and every lab still works.

Open: http://localhost:8000/#/integrations (Integrations > AI providers)

Check: You can say which brain the multi-agent workflow will use: codex, claude, openai or the rule-based fallback.

Step 4. Read the market clock

Look at the US session badge in the top bar. Singapore is 12-13 hours ahead of New York, so a 09:30-17:30 SGT class sits in the US overnight and pre-market sessions.

Open: http://localhost:8000/#/trading (Paper Trading > clock)

Check: You can explain why orders placed during class become extended-hours LIMIT orders.

Step 5. Tour the Pipeline Map

Open the Pipeline Map and hover each of the 14 stages. Click 'Data ingestion' to see its recent activity.

Open: http://localhost:8000/#/pipeline (Pipeline Map)

Check: Data ingestion and Backtesting are green; you can name the stage that enforces the kill switch.

Expected in the app: A working app with demo data loaded, a verified Alpaca paper account or the simulator, and a known AI-agent brain.

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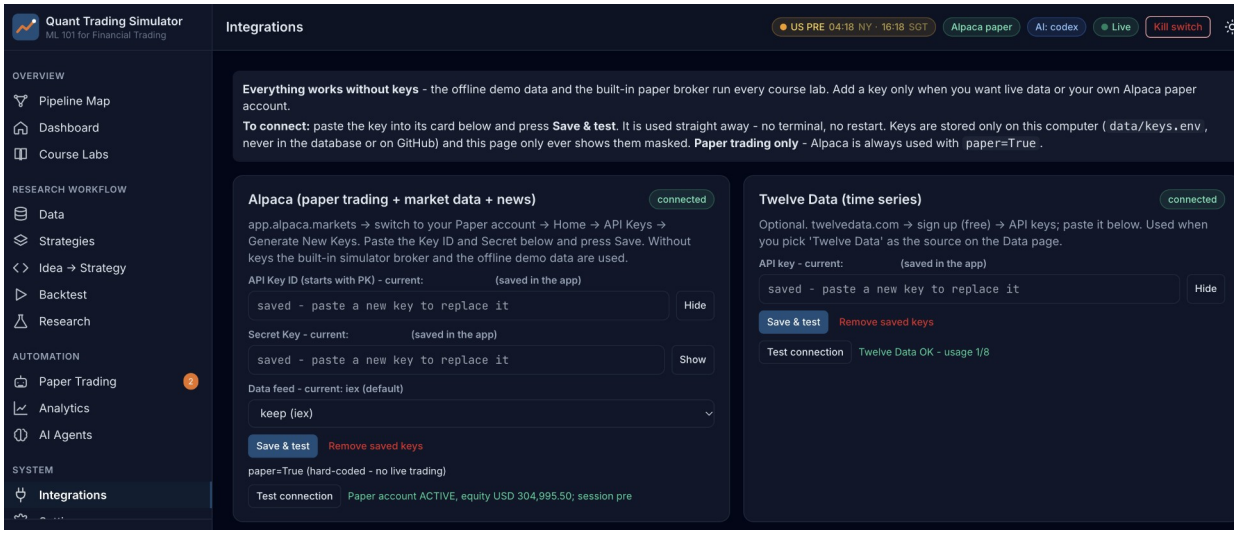


Figure: Lab 01 in the Trading Simulator.

Optional: Let AI Write the Code

This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs/Prompt_Library.html and .pdf.

Market sessions and your first data - run it in Google Colab
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: show the current US market session (pre-market, regular, after-hours, overnight or closed) in New York time and the matching Singapore time using zoneinfo; then download SPY daily bars for 2018-01-01 to 2025-12-31 and print the first and last 5 rows and the number of bars.

Open in Claude | Open in ChatGPT | Open Google Colab

Guide: Download the Course Materials: Learner Guide, Slides and Lab Manual

Get your own copy of the courseware before Day 1. Everything is free to download.

1. Go to the course portal

Open lms-tms.tertiaryinfotech.com in your browser (the Tertiary Infotech Training Management System).

2. Request a one-time password

Under Sign in to the TMS, enter the email you registered for the course with and click Send OTP. (Prefer a password? Click Login with password instead.)

3. Key in the OTP

Open the email from the TMS, copy the one-time password and key it in. No OTP, or the class is missing after sign-in? Use Register with your Course Run ID (from your trainer or course administrator).

4. Download the Learner Guide and slides

Open this course. Under Courseware, open the Learner Slides and the Learner Guide and download both PDFs (they open in Google Drive: File > Download).

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5. Download the lab manual

Open Activities / Lab. It holds the labs folder: every lab has its own folder with a LabNN_Instructions.pdf (the lab manual for that lab). Download the whole folder (right-click > Download).

6. Or download everything from GitHub

Open https://github.com/tertiarycourses/C728-AI-Powered-Quantitative-Trading-Fundamentals, click Code > Download ZIP and unzip it. The courseware/ folder holds the Learner Guide (PDF, Word and Markdown), the slides and the Lesson Plan; the labs/ folder holds all twenty lab manuals.

7. Keep the materials together

Unzip into one folder on your laptop, for example Documents/C728-Quant-Trading, so the Learner Guide, the lab manuals and the simulator sit side by side.

Each lab folder has its own LabNN_Instructions.pdf - the step-by-step lab manual.

Guide: Download, Install and Start the Trading Simulator Software

The labs run in the course's trading simulator - a user-friendly web app on your own laptop. Install it once before Lab 1. Choose Option A (uv) or Option B (Docker). It is a one-time software install - no coding; your trainer can help.

1. Install the prerequisites

Install Git and Python 3.11 or later. Then install uv: on macOS or Linux run curl -LsSf https://astral.sh/uv/install.sh | sh ; on Windows (PowerShell) run powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1 | iex". For Option B install Docker Desktop instead.

2. Download the software from GitHub

Open https://github.com/alfredang/algorithmic-trader-bot, click Code > Download ZIP and unzip it into your course folder.

3. Option A - install and start with uv

Run: uv sync (installs everything), then uv run trader seed (loads the demo data, about 15 seconds), then uv run trader serve --with-worker (web app, API and the automation scheduler in one process). Leave the terminal open.

4. Option B - start with Docker Compose

Run: docker compose up -d --build and wait until all four containers are healthy (docker compose ps). Load the demo data with docker compose exec backend trader seed . Stop it later with docker compose down (your data is kept).

5. Open the app

Open http://localhost:8000 in your browser (if port 8000 is busy: uv run trader serve --with-worker --port 8010). The onboarding wizard lets you choose the offline demo or connect your Alpaca paper account; Settings > Load demo data reloads the demo at any time.

6. Connect Alpaca and the AI agents in the app

Keys are never typed into a file: open Integrations in the app to paste your Alpaca paper keys (next guide) and, optionally, to see the AI agent cards. Without AI logins the multi-agent workflow runs its rule-based agents - every lab still works.

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7. Start Lab 1

In the app's sidebar open Course Labs > Lab 1 and follow the checklist alongside Lab01_Instructions.pdf. If the README of the software gives a different command or port, follow the README - it is the authoritative start-up guide.

Simulator: github.com/alfredang/algorithmic-trader-bot - the README has the full start-up steps and troubleshooting.

Guide: Sign Up for Alpaca and Open the Paper-Trading Account

Alpaca paper trading is free and needs no deposit. You use it in every lab that places orders. Do this before Lab 1.

1. Create the account

Go to alpaca.markets and choose Sign Up. Register with your email address and a strong password, then open the verification email and confirm your address.

2. Log in to the dashboard

Log in at app.alpaca.markets and go to the Home page.

3. Switch to the paper account

In the upper-left corner choose your 'Paper Trading' account. Paper trading needs no deposit and no brokerage approval; every order uses virtual money.

4. Generate your paper API keys

On the paper account's Home page find the API section and generate a key. Copy the Key ID and the Secret Key immediately - the secret is shown only once. Generating a new key invalidates the old one.

5. Store the keys in the simulator

In the Trading Simulator open Integrations > Alpaca, paste the Key ID and the Secret Key and press Save & test. The keys stay on your laptop and are only ever shown masked. Never put keys in a slide, a chat, an AI prompt or any file you share.

6. Check the connection

On the same card press Test connection: it must report 'Paper account ACTIVE' with your equity. Paper keys only work on the paper endpoint (paper-api.alpaca.markets) - they cannot touch a live account.

7. If you cannot open an account

Skip the keys. Every lab runs OFFLINE on the simulator's demo data and its built-in paper broker, so nothing in the course is blocked.

No deposit needed. Paper keys only work on paper-api.alpaca.markets. No account? Every lab also runs offline.

Debrief

A trading pipeline is only as safe as its configuration and its connection to the broker. One validated config file, paper-only enforcement, keys held in the environment and a pre-flight check that a scheduler can read from the exit code show how AI and algorithmic trading programs are connected to execution platforms in a controlled way.

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